

DAYANANDA SAGAR COLLEGE OF ENGINEERING

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning**



INTERNSHIP REPORT

ON

“MarTech-Web”

Submitted in partial fulfillment for the award of degree(20AI8CINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Ch Naganoshith- 1DS20AI020

Conducted at



Alcon

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “**MarTech-Web**” carried out by **Mr. CH NAGANOSHITH [1DS20AI020]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (**20AI8CINT**)

Signature of Reporting Head

Prashant Aggarwal
Solution Delivery
Manager, Alcon

Signature of Internship Coordinator

Prof. Kavya D N
Dept. of AI & ML DSCE

Signature of HoD
Dr. Vindhya P Malagi

Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **CH NAGANOSHITH**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **ALCON**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

USN: 1DS20AI020

Place: Bangalore

NAME: CH NAGANOSHITH

OFFER LETTER



Date: 5 December 2023

Ch Naganoshith
Dayananda Sagar College of Engineering
Shavige Malleswaram Hills, 91st Main Rd,
1st Stage, Kumaraswamy Layout,
Bengaluru, Karnataka 560078

AGREEMENT

SUB: Internship with Alcon Laboratories (India) Pvt. Ltd.

We would like to congratulate you on being selected for an internship with **Alcon** based at **Bangalore**.

Your internship is scheduled to start effective **1 Feb, 2024** for a period of **6 months** as per the internship request letter issued by **Dayananda Sagar College of Engineering** with letter dated **4 Dec 2023**.

You should report for the internship at the following address:

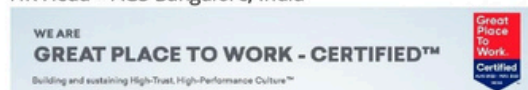
Alcon Laboratories (India) Pvt. Ltd.
Godrej Woodsman Estate RMZ Azure (IT Complex),
9th, 10th & 11th, Bellary Rd, Hebbal,
Bengaluru, Karnataka 560092

Reporting Time: 10:00 AM
Contact Person: Santhosh Kumar SG
Department: Human Resources (Talent Acquisition)

You will be eligible to receive a **stipend of INR 25,000 per month** during your internship. Other terms are mentioned in Annexure enclosed with this letter.

We look forward to welcome you and wishing you an enriching learning experience with us.

Sincerely,
Indumathy Karthikeyan
HR Head – AGS Bangalore, India



Acknowledgement:

I hereby acknowledge, confirm, and agree to the contents of this letter including the Annexure.

Ch Naganoshith

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Mr. Prashant Aggarwal, Alcon, for guiding me during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

CH NAGANOSHITH [1DS20AI020]

TABLE OF CONTENTS

Sl no	Description	Page no
1	INTRODUCTION	1
2	DESCRIPTION OF ORGANIZATION	2
3	SCOPE OF WORK LEARNING	3-4
4	OBJECTIVES	5-6
5	CHALLENGES AND SOLUTIONS	6-7
6	ACHIEVEMENTS	8-9
7	REFLECTION AND ANALYSIS	10-11
8	RECOMMENDATIONS	12
	CONCLUSION	13
	REFERENCES	14

CHAPTER 1

INTRODUCTION

In the realm of cutting-edge eye care solutions, Alcon stands as a beacon of innovation and excellence, dedicated to enhancing vision health worldwide. With over 75 years of pioneering heritage, Alcon has solidified its position as the largest eye care device company globally, serving patients in over 140 countries. Their commitment to helping people see brilliantly is unwavering, driving every aspect of their operations, from research and development to manufacturing and distribution.

On another frontier of innovation, the MarTech team, within its dynamic and innovative domain, is poised to revolutionize webpage management. As an intern within this visionary group, the journey extends far beyond conventional website construction. It encompasses a holistic approach, integrating state-of-the-art technologies, rigorous quality assurance processes, and strategic marketing initiatives to deliver unparalleled digital experiences for clients and end-users worldwide.

In this report, we delve into the symbiotic relationship between Alcon's mission of enhancing vision health and Martech's dedication to crafting exceptional digital experiences. By merging the relentless pursuit of excellence from Alcon's legacy with the forward-thinking ethos of the Martech team, we explore how these two entities converge to push the boundaries of innovation in eye care and digital technology. Through meticulous website development, robust quality assurance measures, and strategic marketing endeavors powered by Salesforce Marketing Cloud and Customer Data Platform solutions, we unveil a narrative of innovation, creativity, and client success.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Alcon is the global leader in eye care, dedicated to helping people see brilliantly. With an over 75- year heritage, they are the largest eye care device company in the world, with complementary businesses in Surgical and Vision Care. Being a truly global company, they work in over 50 countries and serve patients in more than 140 countries. Alcon is grounded in cutting-edge innovation and breakthrough technology, transforming the way eye diseases and eye conditions are treated. They have the widest array of eye care offerings in the industry with products organized into two business: Surgical and Vision Care.

Alcon's vision is to pioneer advancements in eye care globally, setting new standards of excellence and innovation. They aspire to lead the industry by delivering transformative solutions that improve vision health and enhance the lives of individuals worldwide. With a commitment to pushing the boundaries of what's possible, Alcon aims to be at the forefront of research, development, and delivery of cutting-edge eye care products and services.

Their vision encompasses a dedication to addressing unmet needs, expanding access to quality eye care, and ultimately, empowering people to experience the world with clarity and confidence. By striving for excellence in every endeavor, Alcon aims to redefine the future of vision health and make a meaningful difference in the lives of patients and communities around the globe.

The mission of Alcon is to improve vision health worldwide by relentlessly pursuing the goal of helping people see brilliantly. This mission is reflected in every aspect of their operations, from research and development to manufacturing and distribution. Their commitment to quality, safety, and corporate responsibility underscores their mission to make a positive impact on eye health globally. Through their extensive portfolio of products and solutions, Alcon aims to address the diverse needs and challenges faced by patients and healthcare professionals alike, ensuring that everyone has the opportunity to enjoy clear vision and optimal eye health.

CHAPTER 3

SCOPE OF WORK

As adept Drupal developers, our mastery of PHP for server-side validation, coupled with our fluency in HTML, CSS, and JavaScript, forms the cornerstone of our skill set. These languages collectively empower us to craft dynamic and secure websites tailored precisely to our clients' needs.

Harnessing the formidable capabilities of Drupal, we leverage its flexibility and scalability to seamlessly integrate bespoke functionalities and third-party extensions, ensuring each website fulfills its unique project requirements. Our project management endeavors are fortified by Jira, furnishing us with a robust framework for organizing tasks, tracking progress, and fostering efficient collaboration among team members.

With its intuitive interface and comprehensive feature suite, Jira facilitates seamless communication and coordination, thereby facilitating the timely and budget-conscious delivery of projects. Meanwhile, Acquia Cloud stands as our trusted repository for data storage and retrieval, furnishing a secure and reliable infrastructure for hosting our Drupal websites. Its scalability and performance optimization features ensure our websites remain responsive and accessible even amidst substantial traffic influxes.

To uphold stringent quality standards, we employ Percy for rigorous quality assurance testing, automating visual testing across diverse devices and browsers. By proactively identifying visual regressions during the development phase, Percy aids us in upholding the pinnacle of quality and ensuring our websites consistently deliver exceptional user experiences.

In summary, our proficiency in PHP and frontend technologies, combined with the seamless integration of Drupal and complementary tools such as Jira, Acquia Cloud, and Percy, empowers us to deliver robust and innovative web solutions with remarkable efficiency. Through strategic tool utilization and an unwavering commitment to excellence, we continually produce digital experiences that transcend our clients' expectations.

Tools and Technologies used by the MarTech team are:

1. DRUPAL

Alcon employs Drupal as its primary web platform to manage its online presence effectively. With Drupal's versatile content management capabilities, Alcon can create and maintain various components of its digital ecosystem, including its corporate website, product catalog, and educational resources. Drupal's scalability enables Alcon to adapt and expand its online offerings as needed, whether it's integrating e-commerce functionality to facilitate online sales or implementing community engagement features to foster interaction among users. By leveraging Drupal's user-friendly interface and robust features, Alcon can ensure a seamless and engaging experience for visitors, promoting brand awareness, enhancing customer satisfaction, and driving business growth in the competitive landscape of eye care.

2. JAVASCRIPT

In Alcon's web team, JavaScript is a fundamental tool used to enhance the functionality and interactivity of their digital assets. JavaScript enables the team to create dynamic and engaging user experiences on Alcon's websites by incorporating features such as interactive elements, form validation, and dynamic content loading. Additionally, JavaScript facilitates seamless integration with third-party services and APIs, allowing Alcon to leverage the capabilities of external platforms for enhanced functionality and data exchange. JavaScript's AJAX capabilities enable the team to implement efficient and responsive web applications, ensuring smooth user interactions and content delivery. Furthermore, JavaScript plays a crucial role in tracking user behavior and interactions on Alcon's websites, providing valuable insights for analytics and optimization purposes. Overall, JavaScript is an integral part of Alcon's web development toolkit, empowering the team to deliver exceptional digital experiences to their audience.

3. HTML AND CSS

HTML and CSS are foundational languages used to create and style web pages, ensuring a visually appealing and user-friendly experience. HTML (Hypertext Markup Language) provides the structure and content of a webpage, defining elements such as headings, paragraphs, images, and links. CSS (Cascading Style Sheets) complements HTML by controlling the presentation and layout of these elements, including aspects such as colors, fonts, spacing, and positioning.

4. CLOUDFLARE

In the web team, Cloudflare is a vital tool used to enhance website security, performance, and reliability. By leveraging Cloudflare's robust network, the team can mitigate various cyber threats and attacks, ensuring the safety of their websites, APIs, and corporate networks. Cloudflare's advanced security features, such as DDoS protection and Web Application Firewall (WAF), help defend against malicious activities and unauthorized access attempts, safeguarding sensitive data and user privacy. Additionally, Cloudflare accelerates website traffic through its content delivery network (CDN), optimizing page load times and improving overall user experience. The team also utilizes Cloudflare for managing DNS records efficiently, ensuring seamless domain management and propagation. Moreover, Cloudflare's serverless computing capabilities enable the deployment of serverless code, allowing the team to execute functions without the need for traditional server infrastructure. Overall, Cloudflare empowers the web team to deliver safer, faster, and more reliable web experiences to their audience.

5. ACQUIA CLOUD SITE FACTORY

In the web team, Acquia Cloud Site Factory (ACSF) serves as a centralized platform for efficiently managing and scaling a portfolio of websites. The team leverages ACSF to streamline the process of building, managing, and updating multiple websites from a single interface. With ACSF, they can easily deploy new sites and templates, assign roles and permissions, and enforce policies and governance across all sites in the portfolio. ACSF's staging capabilities enable the team to test changes and updates in a controlled environment before deploying them to production, ensuring seamless website management and minimizing the risk of errors or disruptions. Additionally, ACSF provides access to Acquia's team of Drupal experts, offering valuable support and guidance to the web team in optimizing their Drupal-based websites and resolving technical issues efficiently. Overall, ACSF enables the web team to achieve greater efficiency, consistency, and cost-effectiveness in managing their portfolio of websites.

6. NEW RELIC

New Relic serves as a crucial application monitoring tool that provides real-time insights into the performance and health of their web applications. By leveraging New Relic's monitoring capabilities, the team can proactively identify and address any incidents or errors that may occur at various stages of the application lifecycle.

7. BITSIGHT

In the web team, BitSight serves as a crucial tool for assessing and managing cyber risk associated with the organization's digital assets, including websites and online platforms. The team utilizes BitSight's security ratings to gain insights into the security posture of their web infrastructure and areas of improvement.

By regularly monitoring BitSight's cyber risk ratings, the web team can proactively address security issues and mitigate risks before they escalate into major threats. This includes implementing security patches, updates, and best practices to strengthen the resilience of their web assets against cyberattacks. Additionally, BitSight provides the web team with real-time information and business insights, enabling them to make informed decisions about prioritizing security measures and allocating resources effectively. By leveraging BitSight's comprehensive analytics and reporting capabilities, the web team can track their security performance over time and demonstrate compliance with industry standards and regulations. Overall, BitSight empowers the web team to enhance the security and resilience of their digital infrastructure, safeguarding against cyber threats and ensuring the integrity and availability of their web assets.

8. BROWSERSTACK

In the web team, BrowserStack plays a vital role in ensuring the quality and compatibility of their websites across different browsers and devices. By providing developers with a comprehensive testing platform, BrowserStack enables them to perform thorough testing their web applications in real-world environments.

The team utilizes BrowserStack's various testing categories, including Live, Automate, Percy, and Accessibility Testing, to cover different aspects of their testing requirements. With Live testing, developers can interactively test their websites across multiple browsers and devices in real-time, ensuring great accuracy and immediate feedback on any issues encountered.

CHAPTER 5

CHALLENGES AND SOLUTIONS

1. Define Goals and Objectives: Identifying clear goals for the website redesign can be challenging, especially when aligning them with improving usability, increasing engagement, and enhancing accessibility compliance. Conduct thorough stakeholder consultations and research to define specific, measurable, achievable, relevant, and time-bound (SMART) objectives. This ensures alignment with overall business objectives while addressing user needs effectively.

2. Conduct Audience Research: Understanding the preferences and needs of the target audience requires comprehensive research and analysis, which can be time-consuming and resource-intensive. Utilize a combination of quantitative and qualitative research methods, such as surveys, interviews, and user testing, to gain insights into user behavior, preferences, and pain points. Leverage analytics tools to gather data on user interactions with the current website.

3. Assess Current Website: Evaluating the existing website's layout, navigation, content, and accessibility features may uncover numerous areas for improvement, posing challenges in prioritization. Conduct a thorough website audit, involving stakeholders from various departments to identify key pain points and prioritize areas for improvement based on impact and feasibility. Use accessibility auditing tools to assess compliance with accessibility standards.

4. Plan Website Redesign: Developing a comprehensive plan for redesigning the website requires coordination across multiple teams and ensuring alignment with business objectives and user needs. Create a detailed project plan outlining the timeline, tasks, responsibilities, and milestones for the website redesign. Foster collaboration between design, development, content, and marketing teams to ensure a cohesive approach.

5. Update Content Management Strategy: Reviewing and updating the content management strategy involves ensuring regular updates, accuracy, relevance, and optimization for search engines, which can be time-consuming. Implement content management systems (CMS) with robust editorial workflows, content scheduling capabilities, and SEO optimization tools. Provide training and resources to content creators to maintain consistency and quality.

6. Develop Product Showcase: Creating dedicated product pages with detailed descriptions, specifications, and images requires coordination with product teams and integrating interactive features for user engagement. Collaborate closely with product managers to gather comprehensive product information and assets. Implement interactive features such as product comparison tools and virtual try-on experiences to enhance user engagement and decision-making.

7. Optimize User Experience: Conducting usability testing and optimizing user experience requires identifying pain points and opportunities for improvement, which may vary among different user segments. Conduct usability testing with representative user groups to identify usability issues and gather qualitative feedback. Use heatmaps and user journey analysis to optimize navigation, page layout, and conversion paths for a seamless browsing experience.

8. Enhance Accessibility Compliance: Implementing accessibility features such as keyboard navigation and optimized color contrast requires adherence to industry standards and legal requirements, which can be complex. Follow Web Content Accessibility Guidelines (WCAG) and consult accessibility experts to ensure compliance. Use automated accessibility testing tools to identify and remediate accessibility issues efficiently.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

During my internship tenure, I have made significant achievements in redesigning and managing the Alcon website. By focusing on content management, product showcase, user experience optimization, and accessibility compliance, I have enhanced the overall experience for users. The modernized layout, improved navigation, and visually appealing design have made it easier for users to find relevant information quickly and intuitively. Regular updates to the content ensure that users have access to the latest information about Alcon's products and services, thereby increasing engagement and satisfaction.

My efforts in user experience optimization, such as streamlined navigation and optimized page load times, have contributed to a seamless and enjoyable browsing experience, encouraging users to explore the website further. Additionally, implementing accessibility features has ensured that the website is accessible to all users, promoting inclusivity and equal access to information and services.

The incorporation of interactive features such as product comparison tools and virtual try-on experiences has further enhanced user engagement, driving increased interaction and time spent on the site. Overall, these achievements have resulted in a valuable and satisfying online experience for users, ultimately benefiting the Alcon website's audience.

CHAPTER 7

REFLECTION AND ANALYSIS

During my internship at Alcon Eye Care, I collaborated with a dynamic team led by Solution Delivery Manager Prashant Agarwal to develop and maintain the company's websites. Through dedication, creativity, and effective communication, we navigated challenges and delivered quality results. I contributed to tasks such as website maintenance, content management, and visual testing using tools like Percy. Immersed in Marketing technologies (MARTECH), I gained hands-on experience with content management systems, Customer Data Platforms (CDP), and analytics tools, enabling data-driven decision-making.

I not only gained technical skills but also learned valuable lessons about ethical behavior in the corporate world. Under the leadership of Solution Delivery Manager Prashant Agarwal, I witnessed firsthand the importance of integrity, transparency, and accountability in all aspects of business operations. Through open communication and mutual respect, I learned to navigate complex situations with honesty and integrity, always prioritizing ethical considerations in decision-making. As I transition into my future career, I carry with me the invaluable lesson that success is not only measured by outcomes but also by the ethical principles that guide our actions. The final outcome of the internship project focused on redesigning and managing the Alcon website with a comprehensive approach to content management, product showcase, user experience optimization, and accessibility compliance. By prioritizing the needs and preferences of users, the project resulted in an improved website that offers enhanced usability, relevant content, detailed product information, and an engaging user experience. The implementation of accessibility features ensures inclusivity and equal access to information for all users, including those with disabilities. Overall, the internship project successfully delivers a valuable and satisfying online experience for the final users of the Alcon website, contributing to the company's mission of improving vision health worldwide.

CHAPTER 8

RECOMMENDATIONS

Given the open and approachable nature of leaders and mentors at Alcon, along with the fostering and supportive company culture, the internship program can be further enhanced by implementing the following recommendations:

Facilitating Cross-Departmental Collaboration:

Encourage interns to collaborate across various departments and teams within the organization. Leaders and mentors should facilitate introductions and connections, providing interns with insights into different aspects of the business and expanding their professional network. Foster a culture of peer learning and knowledge sharing through regular meetups, brainstorming sessions, or knowledge-sharing events. Interns should be encouraged to share their experiences, insights, and skills with each other, creating a supportive community of learners.

Offering Flexibility in Project Assignments:

Provide interns with the flexibility to choose or customize their project assignments based on their interests, skills, and career goals. Empower them to pursue projects that align with their passions and aspirations, fostering a sense of ownership and investment in their work.

By implementing these recommendations, Alcon can enrich the internship experience, empowering interns to learn, grow, and thrive in a supportive and inclusive environment while making meaningful contributions to the organization's success.

CHAPTER 9

CONCLUSION

In conclusion, my internship at Alcon Eye Care was a transformative experience where I developed technical skills, learned the importance of ethical behavior in the corporate world, and honed teamwork and problem-solving abilities. Under the leadership of Solution Delivery Manager Prashant Agarwal, I contributed to website development, content management, and visual testing, gaining valuable hands-on experience in Marketing technologies (MARTECH). The internship not only equipped me with practical skills but also instilled in me a commitment to integrity, transparency, and accountability in all aspects of business.

As I move forward in my career, I carry with me the lessons learned and the invaluable experiences gained during this internship. This internship served as a platform for personal and professional growth, allowing me to expand my horizons and develop a deeper understanding of the intersection between technology, business, and ethics. Through collaborative projects and mentorship from seasoned professionals like Prashant Agarwal, I learned to navigate complex challenges with resilience and determination. Additionally, the exposure to Marketing technologies provided me with insights into the evolving landscape of digital marketing and the importance of data-driven decision-making. The emphasis on ethical behavior and integrity has left a lasting impression on me, shaping not only my professional conduct but also my personal values.

Moving forward, I am confident in my ability to contribute effectively to any organization while upholding the highest standards of ethics and integrity. I am grateful for the opportunities and experiences gained during my internship at Alcon Eye Care, and I look forward to applying my newfound skills and knowledge in my future endeavors.

CHAPTER 10

REFERENCES

Name: Prashant Aggarwal

Position: Solution Delivery Manager

Company: Alcon

Contact: prashant-1.aggarwal@alcon.com